XIUYI (ZOE) WU

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EDUCATION

University of Michigan – Ann Arbor, MI, USA 2025.08 – 2027.06 (Expected)
Master of Biomedical Engineering (GPA: 3.95 / 4)

University of Macau, Macau SAR, China 2021.08 – 2025.06
Bachelor of Science in Biomedical Sciences (GPA: 3.69 / 4)

RESEARCH EXPERIENCE & PROJECTS

Botulinum Neurotoxin Modulation of Ankle Impedance During Post-Stroke Walking 2026.05 – now

- Analyzed shank kinematics, acceleration, and joint-torque responses to controlled mechanical perturbations during post-stroke walking using Vicon motion-capture data.
- Processed and standardized gait datasets from multiple sources for machine learning; trained and evaluated models to predict ankle angle and torque, supporting perturbation-response isolation and ankle impedance estimation.

CT-Based Assessment and Anthropometric Prediction of Lower-Extremity Muscle CSA 2026.05 – now

- Quantified lower-extremity muscle cross-sectional area (CSA) from CT images and examined variations in muscle morphology across individuals.
- Developed a regression model to predict maximum muscle CSA from anthropometric measurements.

Manuscript: Wu, X., & Liu, J. *CT-Based Assessment and Anthropometric Prediction of Lower-Extremity Muscle Cross-Sectional Area.* (In preparation).

EOG-Based Gaze-Directed Audio Control System 2026.01 – 2026.04

- Developed and integrated a three-stage analog signal-conditioning circuit for EOG acquisition, incorporating amplification and band-pass filtering.
- Implemented threshold-based gaze classification in LabVIEW to control directional audio in real time; evaluated switching performance and identified signal decay as a limitation to sustained gaze detection.

Optimization of small molecule-induced early cardiac differentiation from hESCs 2024.03 – 2025.06

- Explored serum-free in vitro differentiation strategies to derive cardiomyocytes from human embryonic stem cells (hESCs).
- Performed exploratory experiments to redirect early cardiac progenitor–like cell populations toward pharyngeal arch–derived craniofacial muscle differentiation.

iGEM Biology Competition (Student-led Research Team) 2023

Engineered probiotic oral delivery drug for mitigating gluten intolerance 2023.01 – 2023.10

- Developed a synthetic biology strategy using genetically engineered probiotics to colonize the gut and degrade gluten before immune activation.
- Proposed an "immune-threshold" approach to reduce intestinal gluten immunogen load and attenuate downstream mucosal immune responses.

Novel autophagy repressing CtBP-Trim28 complex and its role in cancer development 2023.04 – 2023.08

- Investigated C2C12 myoblast differentiation toward skeletal muscle and brown adipocyte phenotypes.
- Contributed to high-fat-diet-induced diabetic mouse experiments through blood glucose monitoring, tissue dissection, and histological sample preparation.

SKILLS

Computational & Biomechanical Analysis: Python, MATLAB, R, LabVIEW; statistical modeling, machine learning, kinematic and kinetic data processing, gait data analysis; 3D Slicer, Vicon motion capture (marker labeling and gap filling), OpenSim, 3DSSPP.

Experimental & Laboratory Techniques: Human movement experiments; cell culture, histological and immunofluorescence staining, flow cytometry, DNA/RNA analysis, Western blotting, and quantitative image analysis (ImageJ).

AWARDS & HONORS

UM Dean's Honour List 2021-2025
2024 Summer Programme Sponsorship 2024.10
iGEM 2023 – Silver Award 2023.11
UM RC Foundation Scholarship 2022.11
Well-rounded Development Award Programme 2021/2022 – Silver Award 2022.09